

FPGA Synthesis and Prototyping Lab:

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Encoders/Decoders

Introduction

The purpose of this experiment is to Synthesis and Check the behavior of Encoders/Decoders circuits mentioned in Lab 4, 5, 6

The implementation and verification of this experiment were carried out using the VSD FM FPGA development board available with me. The design flow, synthesis, implementation, and hardware validation were completed successfully on this FPGA platform. I hope this submission will be considered for evaluation.

CODE:

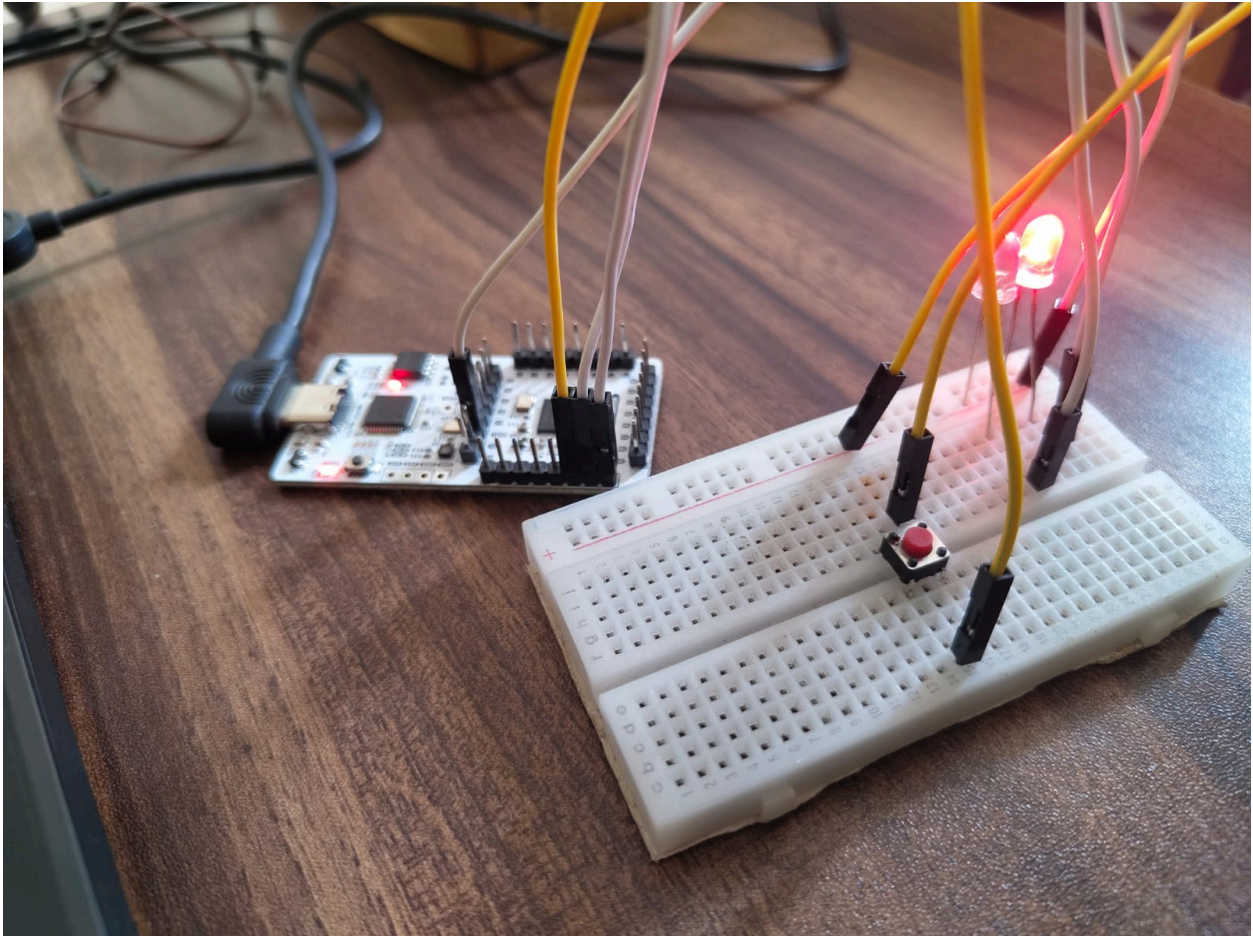
```
module encoder4x2(
    input switchcase,
    output reg [1:0] led
);

reg [3:0] variable1 = 4'b0100;
reg [3:0] variable2 = 4'b0010;

always @(*) begin
    if(switchcase) begin
        led[0] = variable1[1] | variable1[3];
        led[1] = variable1[2] | variable1[3];
    end
    else begin
        led[0] = variable2[1] | variable2[3];
        led[1] = variable2[2] | variable2[3];
    end
end

Endmodule
Constraints
```

```
1 set_io switchcase 35  
2 set_io led[0] 36  
3 set_io led[1] 37
```



Working link
Github [link](#)